

Spring 2025

Math F251X

Calculus I: Final Exam

Name: Solution

Section: ☐ 9:15 (James Gossell)
☐ 11:45 (Kevin Meek)
☐ Online (Deven Barnett)

Rules:

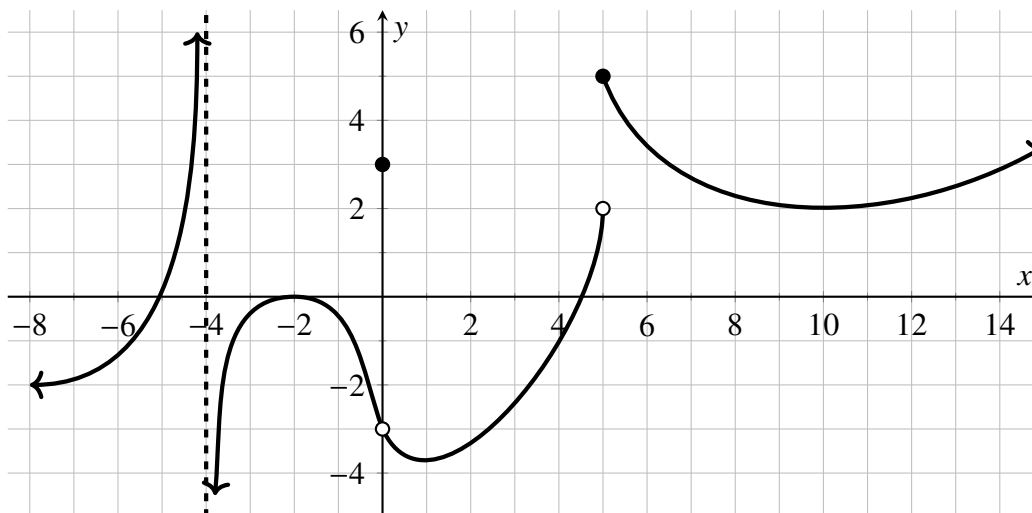
- Partial credit will be awarded, but you must **show your work**.
- You may have a single handwritten $3'' \times 5''$ notecard, both sides. Calculators are **not allowed**.
- Place a box around your **FINAL ANSWER** to each question where appropriate.
- Turn off anything that might go beep during the exam.
- You have two hours to complete the exam.

Problem	Possible	Score
1	15	
2	6	
3	9	
4	10	
5	5	
6	10	
7	12	
8	12	
9	15	
10	6	
Extra Credit	(5)	
Total	100	

11:00

1. (15 points)

Consider the graph of the function $f(x)$ shown below, and answer the following questions. You must give the most complete answer; if the value is infinite, write $+\infty$ or $-\infty$.



a. $\lim_{x \rightarrow -4^+} f(x) = -\infty$ b. $\lim_{x \rightarrow -2^-} f(x) = 0$ c. $\lim_{x \rightarrow 0^-} f(x) = -3$ d. $\lim_{x \rightarrow 5^-} f(x) = \infty$

e. At what x -value(s) is the function $f(x)$ **discontinuous**? $x = -4, 0, 5$

f. At what x -value(s) does $f(x)$ have a **local minimum**? $x = 1, 10$

g. The line $y = -2$ is a horizontal asymptote of $f(x)$. Fill in a statement about a limit that corresponds to this fact.

$$\lim_{x \rightarrow -\infty} f(x) = -2$$

h. On what interval(s) is $f(x)$ **concave up**? $(-\infty, -4) \cup (0, 5) \cup (5, \infty)$

2. (6 points)

Compute the following **derivatives**. Show your work clearly. You do **not** need to simplify your answers. Your answer should start $f'(x) =$, $\frac{df}{dx}$, or similar.

a. $f(x) = 3x^4 \cos\left(\frac{x^2}{5}\right)$

$$f'(x) = 3x^4 \left(-\sin\left(\frac{x^2}{5}\right) \left(\frac{2}{5}x\right) \right) + \cos\left(\frac{x^2}{5}\right) (12x^3)$$

b. $h(x) = 5 \ln(2) + \frac{x^{2.5}}{\ln(4x^3)}$

$$h'(x) = \ln(4x^3) (2.5x^{1.5}) - x^{2.5} \left(\frac{1}{4x^3} 12x^2 \right) \bigg/ (\ln(4x^3))^2$$

3. (9 points)

Find the **absolute maximum** and the **absolute minimum** of the function $f(x) = x^3 - 3x^2$ on the interval $[-1, 4]$. Justify your answer.

$$f'(x) = 3x^2 - 6x \quad 1 \text{ pt}$$

$$0 = 3x^2 - 6x$$

$$0 = 3x(x - 2) \quad 2 \text{ pts}$$

$$x = 0, 2$$

$$f(-1) = -1 - 3 = -4 \quad 1 \text{ pt}$$

$$f(0) = 0$$

$$f(2) = 8 - 3(4) = -4 \quad 1 \text{ pt}$$

$$f(4) = 64 - 3(16) = 16 \quad 3 \text{ pts}$$

Abs. max: 16

abs min: -4

4. (10 points)

A cabbage is launched upward at 40 meters per second from a cannon that sits 100m above the ground. The height (in meters) of the cabbage relative to the ground after t seconds is given by the function

$$h(t) = -5t^2 + 40t + 100$$

- a. Determine the velocity of the cabbage after 0.5 seconds. Include units.

$$v(t) = -10t + 40$$

$$v(0.5) = -10(0.5) + 40 = 35 \text{ m/s}$$

- b. Determine the acceleration of the cabbage at 0.5 seconds. Include units.

$$a(t) = -10$$

$$a(0.5) = -10 \text{ m/s}^2$$

- c. Determine when the cabbage will reach its maximum height. Justify your answer using calculus. Include units.

$$v(t) = 0$$

$$0 = -10t + 40$$

$$t = 4$$

$a(t) < 0$ everywhere
Max at $t = 4$ seconds

5. (5 points)

Let $h(x) = 1 - 3 \tan(x)$

- a. Find the linearization $L(x)$ of $h(x)$ near $x = 0$.

$$L(x) = h(a) + h'(a)(x-a)$$

$$h(0) = 1 - 3 \tan(0) = 1$$

$$h'(x) = -3 \sec^2(x)$$

$$h'(0) = -3$$

- b. Use your answer from part (a) to approximate $\tan(0.1)$.

$$L(x) = 1 - 3x$$

$$h(0.1) \approx 1 - 3(0.1)$$

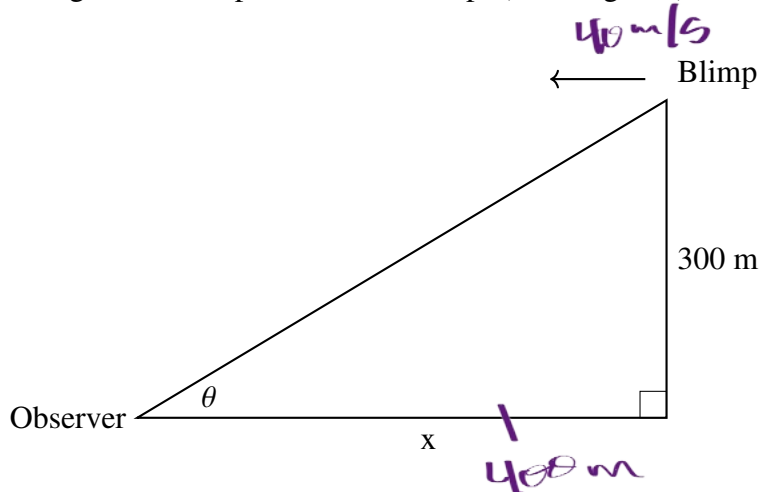
$$= 1 - 0.3$$

$$= 7/10$$

6. (10 points)

Related Rates

A blimp is flying in a straight line at a constant altitude of 300 meters towards an observer who is watching from the ground. The blimp is moving horizontally at 40 meters per second. Let θ be the angle between the ground and the line of sight from the person to the blimp. (See diagram.)



- a. Write an equation that related the variables x and θ .

$$\tan(\theta) = \frac{300 \text{ m}}{x} \quad \text{or} \quad x \tan \theta = 300$$

- b. Determine how fast is the angle of elevation θ changing when the blimp is 400 meters horizontally from the observer? Give units in your final answer.

$$\frac{d}{dt} x \tan \theta = \frac{d}{dt} 300$$

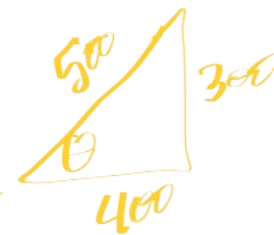
$$x \sec^2 \theta \frac{d\theta}{dt} + \tan \theta \frac{dx}{dt} = 0$$

$$400 \sec^2 \theta \frac{d\theta}{dt} + \tan \theta (-40 \text{ m/s}) = 0$$

Solving for...

Need θ

or triangle



125.98
 $\frac{500}{400} = \frac{5}{4}$
 $\frac{5}{4} = \sec \theta$

$$400 \left(\frac{5}{4} \right)^2 \frac{d\theta}{dt} + \frac{300}{400} (-40 \text{ m/s}) = 0$$

$$625 \frac{d\theta}{dt} = 30 \text{ m/s}$$

$$\frac{d\theta}{dt} = \frac{30}{625} \text{ rad/s}$$

7. (12 points)

Compute the following **limits**. Show your work clearly. Be sure to use **proper notation** where appropriate. Solutions without proper notation will not receive full credit. Indicate any use of L'Hôpital's Rule with $\stackrel{h}{=}$ or $\stackrel{L'H}{=}$ or something similar. (Be sure to verify explicitly that L'Hôpital's Rule applies!).

a. $\lim_{x \rightarrow \infty} \frac{4x}{\sqrt{x^2 - 1}}$

form $\frac{\infty}{\infty}$ can factor in denominator.

$$\lim_{x \rightarrow \infty} \frac{4x}{|x| \sqrt{1 - \frac{1}{x^2}}} = \lim_{x \rightarrow \infty} \frac{4x}{x \sqrt{1 - \frac{1}{x^2}}} = 4$$

b. $\lim_{x \rightarrow \pi} \frac{x - \pi}{\sin(x)}$

form $\frac{0}{0}$

$$\stackrel{L'H}{=} \lim_{x \rightarrow \pi} \frac{1}{\cos(x)} = -1$$

c. $\lim_{x \rightarrow \infty} x^2 e^{-x}$

form $\infty \cdot 0$

$$= \lim_{x \rightarrow \infty} \frac{x^2}{e^x} \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{x}{e^x} \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

d. $\lim_{x \rightarrow 0} \frac{e^x}{\cos(x)}$

form $\frac{1}{1}$

$$= 1$$

8. (12 points)

Compute the following **integrals**. Give the most general answer, and show your work. Clearly indicate any substitutions you use.

a. $\int_0^3 t^2 + 6t - e^t dx$

$$\begin{aligned}
 &= \left. \frac{1}{3}t^3 + \frac{6}{2}t^2 - e^t \right|_0^3 \\
 &= \frac{1}{3}3^3 + 3(3^2) - e^3 - (0 + 0 - 1) \\
 &= 9 + 27 - e^3 + 1 \\
 &= 37 - e^3
 \end{aligned}$$

b. $\int \csc(3\theta - \pi/2) \cot(3\theta - \pi/2) d\theta$

$$u = 3\theta - \pi/2 \quad du = 3 d\theta$$

$$= -\frac{1}{3} \csc(3\theta - \pi/2) + C$$

final version

c. $\int \frac{12e^x}{e^x + 12} dx$

$$u = e^x + 12$$

$$du = e^x dx$$

$$= \int \frac{12}{u} du$$

$$= 12 \ln|u| + C$$

$$= 12 \ln|e^x + 12| + C$$

9. (15 points)

At 2:00 pm ($t = 0$), the main water line in your apartment bursts. The rate at which water is flowing from the break is given by $r(t) = \frac{5t}{1+t^2}$, where r is measured in gallons per hour and t is measured in hours.

- a. Find $r(3)$. Write a sentence that someone who has not taken calculus could understand to explain the meaning of $r(3)$. Include units.

$$r(3) = \frac{5(3)}{1+3^2} = \frac{15}{10} \text{ or } 1.5 \text{ gallons per hour}$$

at 5:00pm, water is flowing at 1.5 gallons per hour.

$t=3$ is 5:00pm

- b. Evaluate $\int_1^2 r(t) dt$.

$$\int_1^2 \frac{5t}{1+t^2} dt$$

$u = 1+t^2$
 $du = 2t dt$

$$\int_2^5 \frac{5}{2} \frac{1}{u} du = \frac{5}{2} \ln|u| \Big|_2^5 = \frac{5}{2} (\ln(5) - \ln(2))$$

- c. Write a sentence that someone who has not taken calculus could understand to explain the meaning of $\int_1^2 r(t) dt$. Include units.

$$= \frac{5}{2} \ln\left(\frac{5}{2}\right) \text{ gallons}$$

between hours 3:00pm and 4:00pm,
 $\frac{5}{2} \ln\left(\frac{5}{2}\right)$ gallons in total flowed through.

- d. At 3:00pm, is the rate of flow increasing, decreasing or staying the same? Explain how you know.

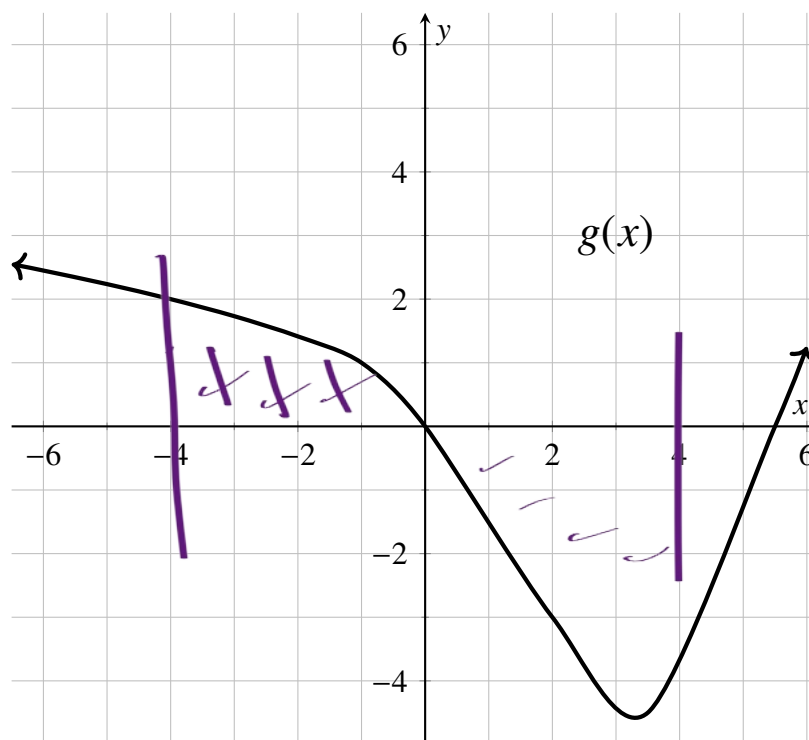
$$r'(t) = \frac{(1+t^2)(5) - 5t(2t)}{(1+t^2)^2}$$

$$r'(1) = \frac{10 - 10}{4} = 0$$

Rate of flow is staying the same.
The change in flow is 0.

10. (6 points)

The graph of a function $g(x)$ is shown below.



Define a new function $A(x) = \int_{-4}^x g(t) dt$.

- a. Determine $A'(2)$.

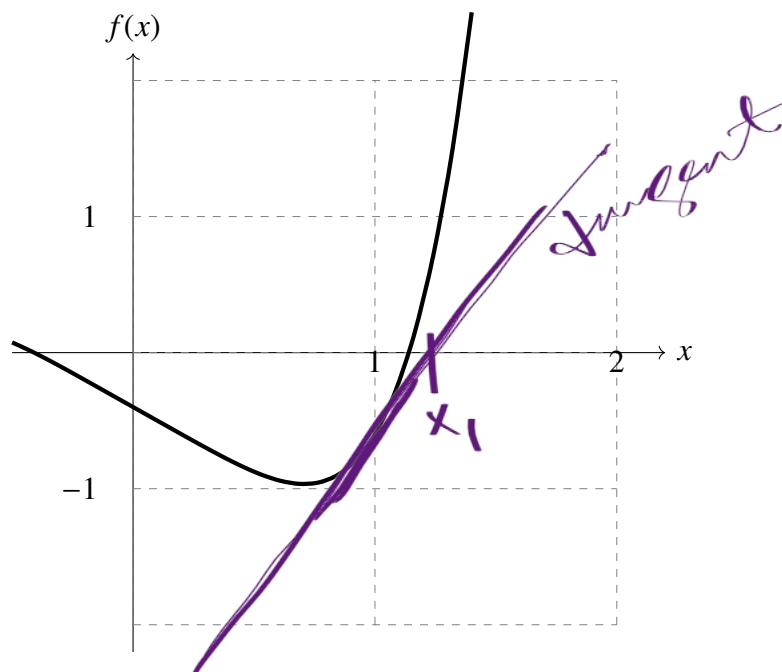
$$A'(x) = g(x)$$

$$A'(2) = g(2) = -3$$

- b. Does $A(x)$ have a local maximum or a local minimum on the interval $(-4, 4)$? If so, give the corresponding x -value and **explain** your answer. If not, explain why not.

$A(x)$ is area under the curve.
 adding area until $x=0$ then subtracting
 area. $A(x)$ has local max
 at $x=0$.
 at $x=1$ updated

Extra Credit (5 points) A portion of the graph of the function $f(x) = \frac{4}{5}x^5 + x - \frac{2}{5}$ is shown below.



- a. Suppose Newton's method is used to find an approximate solution to $f(x) = 0$ from an initial guess of $x_0 = 1$. **Sketch** on the graph how the next approximation x_1 will be found, **labeling** its location on the x -axis.

on graph

- b. If your starting guess is $x_0 = 1$, **compute** x_1 .

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1 - \frac{\frac{4}{5} - 1 - \frac{2}{5}}{4(1)^4 - 1}$$

$$= 1 - \frac{\frac{4}{5} - 1 - \frac{2}{5}}{4 - 1}$$

$$= 1 - \frac{-\frac{3}{5}}{3} = 1 + \frac{1}{5} = \frac{6}{5}$$

$$f(x) = \frac{4}{5}x^5 + x - \frac{2}{5}$$